Regularization's Effect on Bias and Variance

The **regularization parameter (λ)** is a critical tool for controlling the bias-variance trade-off in your model. By adjusting λ, you can directly influence your model's complexity, moving it along the spectrum from underfitting (high bias) to overfitting (high variance).

The Role of λ in Model Fit

We'll use a 4th-order polynomial model (which tends to overfit) to see the effects of λ.

- **When λ is very large (e.g., 10,000):**
  - The cost function places an enormous penalty on all weight parameters (w_j).
  - To minimize the cost, the algorithm is forced to set all w_j parameters to be very close to zero.
  - The model effectively becomes $f(x) \approx b$ (a flat horizontal line).
  - This model is too simple and **underfits** the data.
  - **Diagnosis: High Bias**. (J_{train} will be high, and J_{cv} will also be high).
- **When λ is zero:**
  - The regularization term disappears entirely.
  - The model becomes an unregularized 4th-order polynomial, which will fit the training data perfectly (or near-perfectly) by creating a "wiggly" curve.
  - This model is too complex and **overfits** the data.
  - **Diagnosis: High Variance**. (J_{train} will be low, but J_{cv} will be much higher).
- **When λ is "just right" (an intermediate value):**
  - The model finds a balance. It fits a smooth, reasonable curve that captures the data's trend without fitting the noise.
  - **Diagnosis: Low Bias, Low Variance**. (J_{train} is low, and J_{cv} is also low and close to J_{train}).

Visualizing Error vs. λ

We can diagnose the model by plotting the training and cross-validation error as a function of λ.

- **Horizontal Axis:** The value of λ.
- **Vertical Axis:** The error (J).

Curve Behavior

- **Training Error (J_{train}):**
  - When λ is 0, the model overfits, so J_{train} is **very low**.
  - As λ **increases**, the penalty on the weights gets stronger, forcing the model to be simpler and *worse* at fitting the training data.
  - Therefore, J_{train} **consistently increases as λ increases**.
- **Cross-Validation Error (J_{cv}):**
  - This curve has a **U-shape**.
  - **Region 1: Low λ (e.g., $\lambda = 0$):** The model is overfitting. J_{train} is low, but J_{cv} is **high**. This is the **high variance** region.
  - **Region 2: High λ (e.g., $\lambda = 10,000$):** The model is underfitting. J_{train} is high, and J_{cv} is also **high**. This is the **high bias** region.
  - **The "Sweet Spot":** The minimum of the J_{cv} curve represents the optimal value for λ, which provides the best balance between bias and variance.

> ***Note on Plots:*** *This λ vs. Error plot looks like a "mirror image" of the Model Complexity (d) vs. Error plot.*
>
> - *For **degree d**: High Bias is on the left (simple models), High Variance is on the right (complex models).*
> - *For **lambda** λ: High Variance is on the left (complex models), High Bias is on the right (simple models).*

Choosing the Optimal λ (Model Selection Procedure)

You can use your cross-validation set to automatically find the best value for λ.

1. **Create a list of λ values** to try (e.g., `0`, `0.01`, `0.02`, `0.04`, `0.08`, `0.16`, ..., `10`).
2. **Iterate through the list:** For each value of λ:
   a. Train your model on the **training set** to find the corresponding parameters w, b.
   b. Evaluate the trained model on the **cross-validation set** to get its error, J_{cv}.
3. **Select the best λ:** Choose the value of λ that resulted in the **lowest** J_{cv}.
4. **Report final performance:** Evaluate the model (with your chosen λ) on the separate **test set** (J_{test}) to get an unbiased estimate of its generalization error.